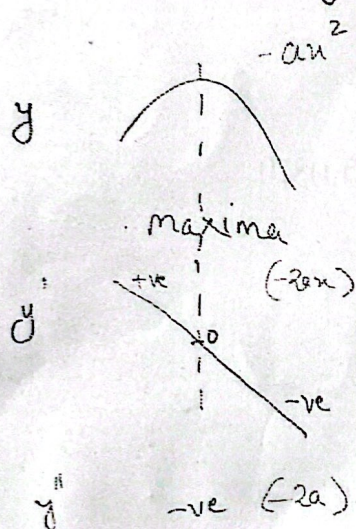
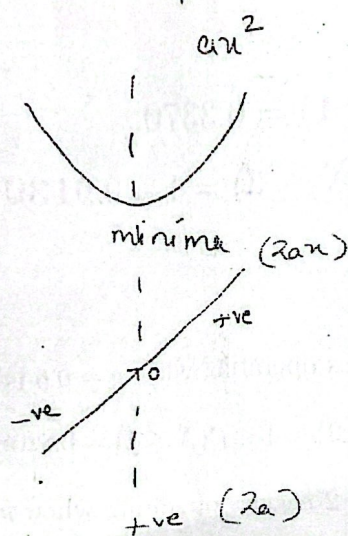


Differentiating maxima v/s minima



Note: $a > 0$

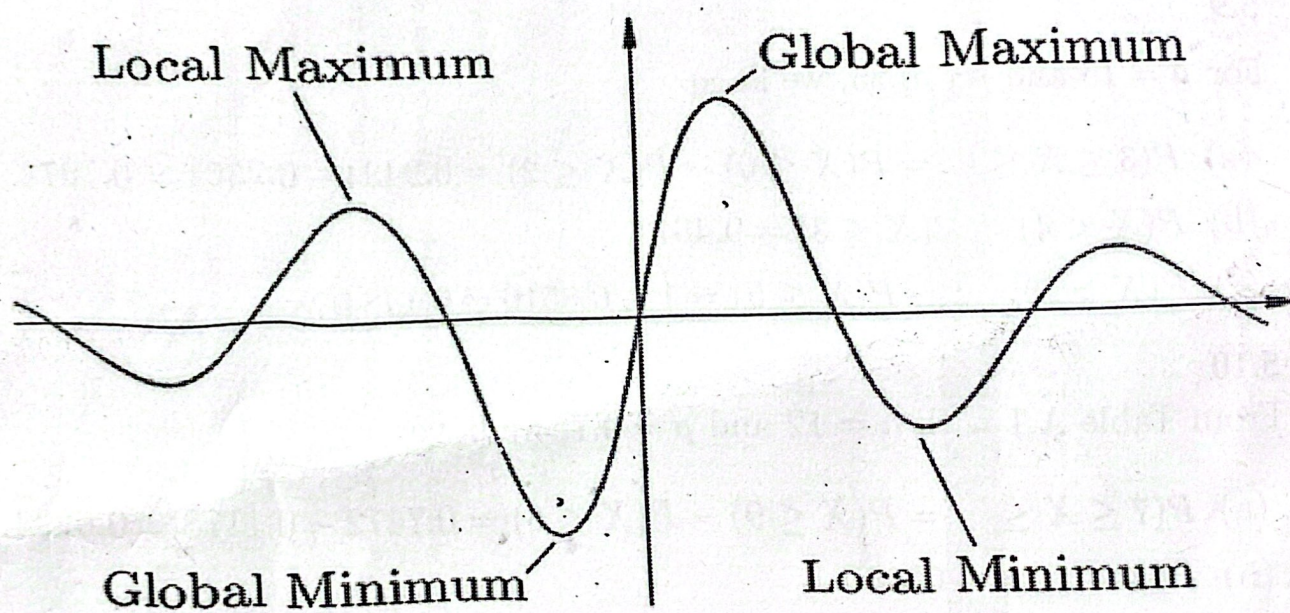


Steps:

- 1) Find critical points
- 2) first derivative test
 - if derivative changes from +ve to -ve at critical point $\Rightarrow$ maxima
 - if derivative changes from -ve to +ve $\Rightarrow$ minima

or
2nd derivative test
if $y'' < 0 \Rightarrow$ maxima
 $y'' > 0 \Rightarrow$ minima

Tip: If there are multiple minima, calculate y values and check which is lowest. That point is global minimum and rest are local minima. 11th test for maximum.



the entire function.

Summary of Differences:

Concept	Local Minimum	Global Minimum	Local Maximum	Global Maximum
Definition	Smallest value within a neighborhood.	Smallest value over the entire domain.	Largest value within a neighborhood.	Largest value over the entire domain.
Scope	Local region around a point.	Entire domain of the function.	Local region around a point.	Entire domain of the function.
Graph	Appears as a "valley" or dip.	Appears as the lowest point of the graph.	Appears as a "peak" or hill.	Appears as the highest point of the graph.
Example	$f(x) = x^2 + 2$ at $x = 0$	$f(x) = x^2$ at $x = 0$	$f(x) = -x^2 + 3$ at $x = 0$	$f(x) = -x^2$ at $x = 0$

In short:

• Ask anything

📎 Attach

🔍 Search

💡 Reason

🗣️ Voice

ChatGPT can make mistakes. Check important info.

Convex Optimization

Convex optimization is a subfield of mathematical optimization that studies the problem of minimizing convex functions over convex sets.

CONVEX SET

If every point on line segment joining any two point, lies on set, then it is called convex set.



Convex Set



Convex Set



Non-Convex Set



convex optimization problems have a single global minimum and are relatively easier

- **Portfolio Optimization:** linear relationship between the expected returns and the risk (covariance) matrix.
- **Least Squares Regression:** In linear regression, the goal is to find the parameters that minimize the sum of squared differences between the observed and predicted values.
- **Support Vector Machines (SVM):** In binary classification, SVM aims to find the hyperplane that maximizes the margin between the two classes while minimizing classification errors.

non-convex optimization problems may have multiple local minima, making them more challenging to solve,

- **Neural Network Training:** Training deep neural networks involves optimizing a non-convex loss function with respect to the model parameters (weights and biases).
- **Image Registration:** In medical imaging, image registration aims to align different imaging modalities or time-series images. The optimization problem involves finding the transformation parameters that minimize the discrepancy between the images.